

Memory Forensics Report — Challenge "Notch It Up" — Executive Summary

CONFIDENTIAL — Forensic Work Product

Single 1.6 GB raw RAM image triaged end-to-end on the live Agentropix-SIFT MCP.
Code injection confirmed across four host processes.

Field	Value		Field	Value
Case ID	CHALLENGE-NOTCHI TUP		Severity	 High
Examiner	victor.galvan		Snapshot (UTC)	2026-06-07T12:40:25Z
Report ID	81a1b2b0...507e5f 91		Generated (UTC)	2026-06-07T12:40:25Z
MCP host	<TAILNET-HOST>		Seal	5 findings · HMAC-SHA256 per finding

Bottom line

A Windows x64 VirtualBox guest RAM image (boot 2019-08-19) shows four PAGE_EXECUTE_READWRITE injected memory regions (MITRE T1055). The decisive indicator is a 64 KB RWX region in `explorer.exe` (PID 1944) carrying executable indirect-jump shellcode (`41 ba ... 48 ff 20`), carved and hashed (payload sha256 `65196e1a...1726ca6f`). A 512 KB RWX region in the WMI host `WmiPrvSE.exe` (PID 2292) couples injection with WMI tradecraft (T1047). All 97 network sockets are benign browser traffic — no external C2 in this snapshot.

KPIs

KPI	Value	KPI	Value
Approved findings	5	MITRE techniques	2 (T1055, T1047)
Hosts in scope	1	Initial access	Not observed (RAM-only)
IOCs catalogued	5	Attacker dwell	Not derivable (snapshot)

Risk matrix

Impact ↓ / Likelihood →	1 Rare	2 Unlikely	3 Possible	4 Likely	5 Almost certain
5 Severe	.	.	.	.	.
4 Major	.	.	.	● F1	● F4
3 Moderate	.	.	● F2, ● F3	.	.
2 Minor	.	.	.	.	.
1 Negligible	.	● F5	.	.	.

Top findings

1. ● F-NOTCH-001 — explorer.exe RWX executable shellcode (PID 1944) (High) — 64 KB RWX VAD at 0x4320000 with indirect-jump shellcode; confirmed process injection.
2. ● F-NOTCH-004 — WmiPrvSE.exe 512 KB RWX (PID 2292) (High) — large RWX region in the WMI provider host; injection + WMI execution surface.
3. ● F-NOTCH-002 / F-NOTCH-003 — zeroed RWX VADs in explorer.exe and chrome.exe (Medium) — corroborating RWX regions flagged for review.

Headline recommendation

[P1] Isolate the host and dump/analyse the 64 KB executable region in explorer.exe (payload sha256 65196e1a...1726ca6f) and the 512 KB region in WmiPrvSE.exe to identify the

implant and confirm WMI persistence.

Disk recall 72/72 (100%) · Memory recall 108/118 (91.5%) · 4464 tests (canonical, .crew/facts.md).

Evidence gate: PASS (5 findings approved + HMAC-sealed). Full report: see comprehensive report

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